The Word Garden - Evaluating Multimodal Multiplayer Games for Cognitive and Social Skills Development in Dyslexic Children

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Group Members

No	Name	IT No	Component Name
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2.	D.P. Samarasekara	IT22599704	Social Engagement System
3.	S.P. Ranasinghe	IT22597656	Predictive Feedback Framework
4.	H.T.P.D.N.P. Gunasekara	IT22610034	Cognitive Interaction Engine

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1.3 Short introduction of the project

i. **What is The Word Garden?** A cooperative multiplayer literacy game where children with dyslexia learn to read together, not alone.

ii. **Who is it for?** Children aged 8-12 with dyslexia, evaluated with clinical oversight at Lady Ridgeway Hospital and Sirimavo Bandaranaike Specialized Children's Hospital, Colombo.

iii. **What problem does it solve?** 1 in 10 children has dyslexia. Yet every existing digital tool makes them face it alone. Current interventions are single-player, text-heavy, and built around correcting failure after it happens. No tool has ever combined real-time multiplayer, cross-device interaction, voice communication, and anticipatory feedback in a single dyslexia-focused experience. Until now.

iv. **How does it work?** Players explore a shared 3D world on PC and discover letter-containing chests. Each discovery triggers a private letter-tracing task on a paired Android device, keeping individual practice hidden from peers to eliminate performance anxiety. Steam Voice lets teammates coordinate freely without face-to-face pressure. The system guides learners before mistakes happen, not after.

v. **What makes it different?** A direct comparison with established tools including GraphoGame, Lexia Core5, and Nessy Learning confirms that none offer multiplayer, cooperative mechanics, dual-device support, or voice communication. The Word Garden is the first system to combine all four, while also being the only one to deliberately exclude competitive elements such as leaderboards that increase anxiety in struggling readers.

vi. **What will it prove?** That children with dyslexia learn better together. This research will generate the first controlled empirical evidence on whether cooperative, voice-enabled, cross-device gameplay improves literacy skills and social confidence, giving educators, clinicians, and developers the evidence base they have been waiting for.

2. The Problem

Dyslexia affects approximately 5-10% of children worldwide, making it the most prevalent neurodevelopmental learning difficulty. In Sri Lanka, a significant proportion of school-age children present with undiagnosed or unsupported reading difficulties, yet specialist digital interventions remain scarce. Beyond letter decoding challenges, dyslexic children suffer measurable harm to self-confidence, experience heightened anxiety in literacy settings, and frequently withdraw from group learning activities, a cycle that compounds academic disadvantage over time. In the commercialization video, this problem will be illustrated through prevalence infographics, real-world data from partner clinical sites, and brief testimonial scenarios from educational therapists at Lady Ridgeway Hospital and Sirimavo Bandaranaike Specialized Children's Hospital, bringing the human dimension of the problem to life for investors and stakeholders.

Key evidence:

- Globally 5-10% of children have dyslexia; prevalence is consistent across languages and educational systems.
- Dyslexic learners show measurably lower reading self-concept and higher trait anxiety compared to typical peers (Polychroni et al., 2024), creating emotional barriers that outlast any single literacy intervention.
- Existing digital tools are predominantly single-player, focusing on decoding accuracy while reinforcing social isolation and neglecting the psychosocial dimensions of sustained engagement (Tlili et al., 2022).
- Post-error feedback, the dominant design pattern in current tools, can reinforce fear of failure rather than preemptively supporting the learner through anticipatory guidance.
- No controlled study has yet combined synchronous multiplayer mechanics, cross-device multimodal interaction, and voice-enabled peer communication in a game designed specifically for dyslexic children, leaving this high-impact design space entirely unvalidated.

3. The Solution

3.1 How it solves the problem

The game addresses dyslexia's cognitive, emotional, and social barriers simultaneously through four mechanisms. First, multimodal distribution spreads learning across visual (3D game world), auditory (voice and sound cues), and kinesthetic (touch-based letter tracing on mobile) channels, reducing the cognitive overload that written-text-dominant tools impose on dyslexic learners. Second, real-time multiplayer collaboration reframes literacy tasks as shared team challenges, normalizing errors and building social confidence in a low-stakes virtual space. Third, an anticipatory feedback mechanism detects likely incorrect selections before they occur and provides gentle real-time cues (visual, audio, and haptic) that guide the learner proactively rather than correcting after failure. Fourth, voice-enabled peer communication via Steam Overlay allows children to coordinate freely without visual exposure, reducing communication anxiety while fostering natural cooperative interaction.

3.2 Key features of the game

- Real-time multiplayer 3D environment (Unreal Engine) with collaborative word-formation tasks and shared letter-chest discovery.
- Mobile companion app (Flutter/Android) supporting guided letter tracing, reduced-assistance tracing, and recognition-based tasks triggered synchronously by in-game events.
- Anticipatory feedback engine delivering real-time visual, audio, and haptic cues before errors occur to reduce frustration and build proactive confidence.
- Steam Overlay Voice integration for anonymous peer communication, enabling natural cooperation without the social anxiety associated with face-to-face interaction.
- Non-competitive reward system using collective achievement indicators (stars, progress visuals) rather than leaderboards, framing success as a shared accomplishment.
- Firebase Realtime Database middleware for low-latency bidirectional synchronization between PC and mobile devices via session code pairing.
- Adaptive difficulty levels (guided tracing, reduced-assistance tracing, full recognition) with color-profile personalization to accommodate individual learning needs.
- Timestamped event logging of gameplay actions, task completion, and interaction patterns to support rigorous research evaluation.

3.3 Research outcomes

- Empirical evidence comparing multimodal multiplayer gameplay against traditional single-player literacy approaches for dyslexic children aged 8-12.
- Insights into how voice-enabled peer collaboration influences social confidence, communication behavior, and fear of failure in low-pressure virtual environments.
- Analysis of anticipatory feedback effectiveness in reducing frustration and improving task accuracy and learning retention compared to conventional post-error correction.
- Design guidelines for dyslexia-sensitive multiplayer educational games addressing both cognitive processing constraints and socio-emotional needs.
- A validated research prototype demonstrating the feasibility of an integrated two-device,

voice-enabled multiplayer learning game, with a foundation for future clinical and educational trials.

3.4 What makes your idea innovative or different from existing solutions

No existing game combines all four distinguishing elements of this project. Current dyslexia software is universally single-player and single-device, uses reactive post-error feedback, and ignores the social and emotional dimensions of learning. This game is the first to: (1) bridge a PC game environment and a mobile companion application in real time for dyslexia support; (2) integrate voice-enabled anonymous peer communication within a literacy learning game; (3) implement anticipatory rather than retrospective feedback; and (4) measure psychosocial outcomes (confidence, fear of failure, and social participation) alongside literacy performance metrics. The combined novelty of cross-device synchronization, cooperative gameplay, and proactive emotional scaffolding positions this as foundational evidence for next-generation assistive educational technologies.

4. Market Opportunity

The global educational technology (EdTech) market for students with learning disabilities is growing rapidly, driven by increasing awareness of neurodivergent needs and a policy shift toward inclusive education. In Sri Lanka specifically, the adoption of digital learning tools is accelerating in both mainstream schools and specialist facilities such as Sirimavo Bandaranaike Specialized Children's Hospital and Lady Ridgeway Hospital, both of which are project partners. Globally, the assistive education technology segment is projected to exceed USD 4 billion by 2027.

Target audiences include: primary schools and special education units seeking engaging digital interventions; speech-language therapy and child development clinics requiring structured literacy tools; parents and caregivers wanting at-home supplements to formal support; and NGOs and government-led inclusive education programmes in Sri Lanka and the wider South Asian region. A modular licensing model covering institutional subscriptions for schools and clinics, family-tier access, and optional premium content packs provides a clear path to sustainable revenue. The game's low hardware requirements (standard Windows PC and Android device) keep the barrier to adoption low even in resource-constrained settings.

Now is the right time to commercialize this research for three converging reasons. First, Sri Lanka's 2023 National Education Policy explicitly mandates inclusive digital learning tools in state schools, creating an immediate institutional demand that did not exist two years ago. Second, the global EdTech investment climate is actively seeking evidence-based interventions for underserved learner populations, with assistive learning technology attracting record funding rounds in 2023-2024. Third, this project is completing its prototype and clinical evaluation phase in 2025-2026 precisely as both clinical partners begin their digital intervention procurement cycles, meaning a validated product can enter deployment without delay.

5. Competitive Edge

5.1 Advantages

- Only game combining real-time multiplayer gameplay, cross-device synchronization, voice communication, and anticipatory feedback in a single dyslexia-focused design.
- Addresses psychosocial outcomes (confidence, fear of failure, social participation) that competitors ignore, making it holistically more effective for long-term engagement.
- Built on stable, open-source-friendly technologies (Unreal Engine, Flutter, Firebase, Steam SDK) ensuring low development cost and straightforward scalability.
- Designed with clinical partners (Lady Ridgeway Hospital, Sirimavo Bandaranaike Hospital) ensuring ecological validity and a direct pathway to real-world deployment.
- Non-competitive reward design reduces performance anxiety, a deliberate differentiator from gamified EdTech tools that inadvertently increase pressure through leaderboards and scoring.

5.2 Supporting evidence

- Vonthron et al. (2024) demonstrated high usability and preliminary literacy gains with a rhythm-training serious game for dyslexic children, confirming that game-based multimodal approaches are feasible and well-received.
- Vasalou et al. (2017) showed that group game-play for dyslexic children produced spontaneous peer scaffolding and increased social engagement, providing direct evidence that multiplayer contexts benefit this population.
- Tlili et al. (2022) systematic review confirmed that game-based learning benefits learners with disabilities through increased engagement and social participation, while identifying the absence of multiplayer dyslexia-specific studies as a critical gap, which this project directly fills.
- Lee et al. (2018) found that second-screen companion applications improve attention allocation under medium cognitive load, supporting the dual-device architecture adopted by this game.
- The functional prototype itself provides direct evidence of feasibility: both the Unreal Engine PC environment and the Flutter mobile companion application are implemented and synchronized via Firebase, the anticipatory feedback mechanism is operational, and Steam Overlay Voice integration is active. Expert walkthroughs with supervisors and clinicians at Lady Ridgeway Hospital have confirmed the prototype is functional, accessible, and appropriate for the target age group, positioning this project well ahead of most academic-stage competitors who remain at conceptual or wireframe level.

- A direct feature comparison with established dyslexia interventions including GraphoGame, Lexia Core5, and Nessy Learning confirms that none of these platforms offer multiplayer, cooperative mechanics, dual-device support, or voice communication, all of which are core features of this system. Unlike platforms that use leaderboards which may heighten anxiety in struggling readers, this project deliberately excludes competitive elements, positioning it as complementary to validated phonics-based tools rather than a replacement.

6. Development and Implementation

6.1 Initial Sketches

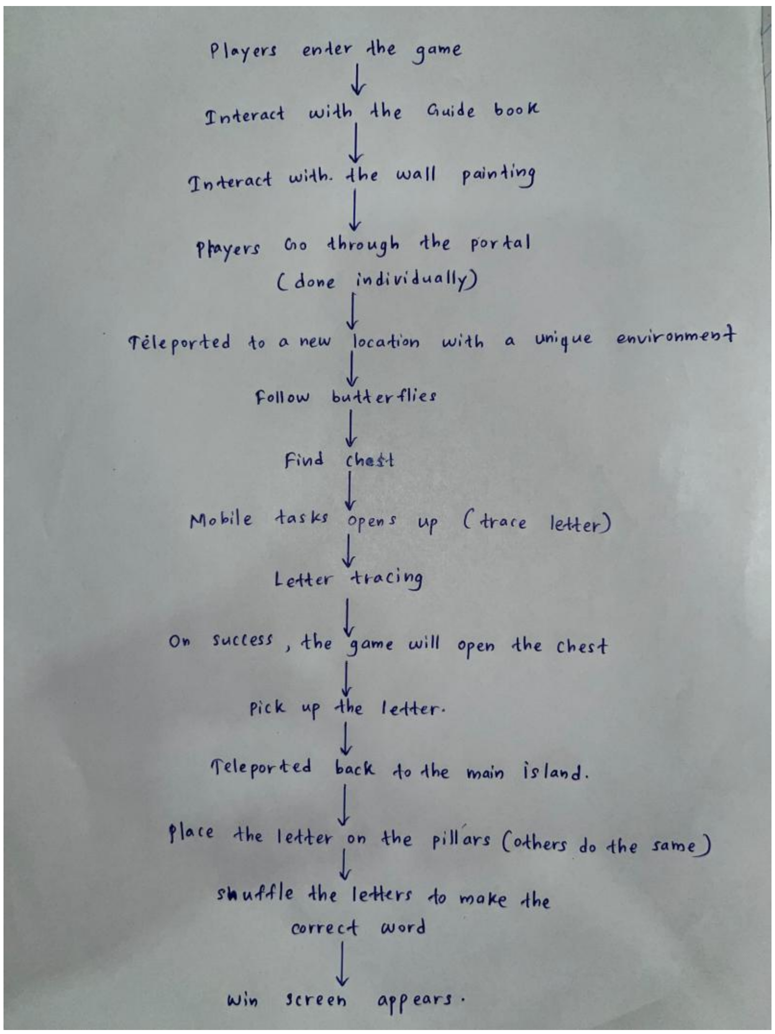

Figure 1: The Session Pairing Flow

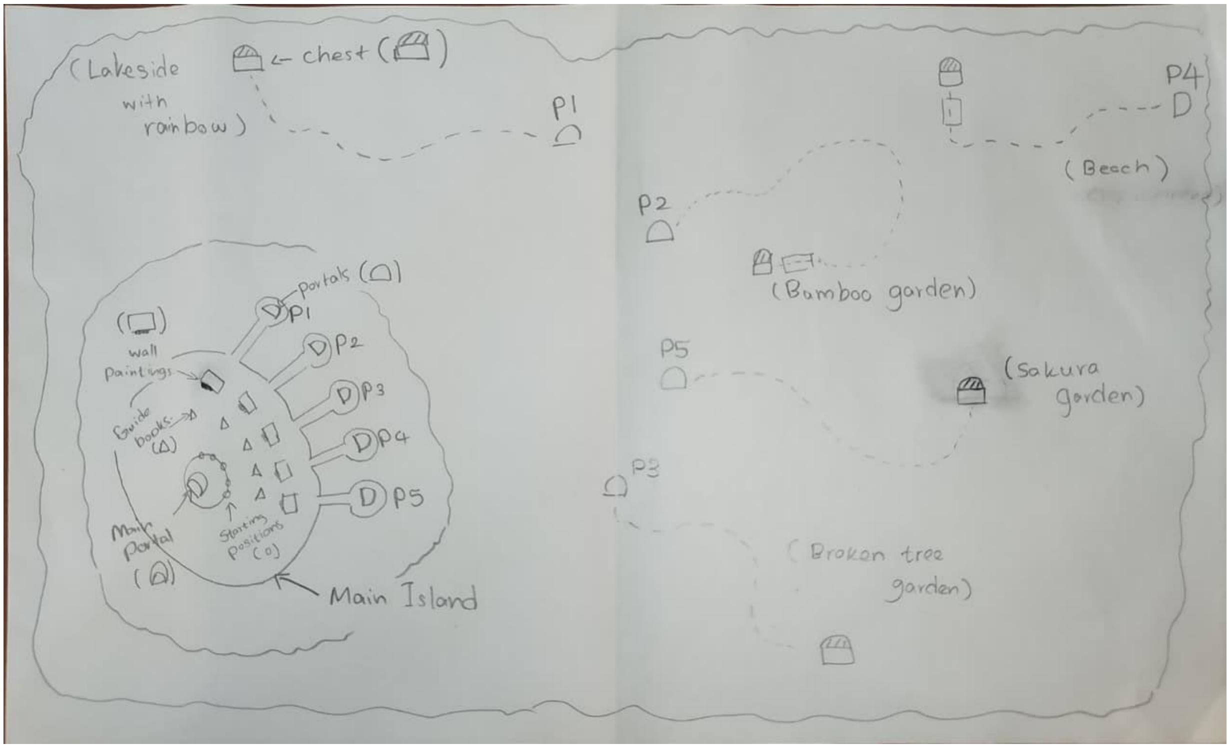

Figure 2: The PC game world layout

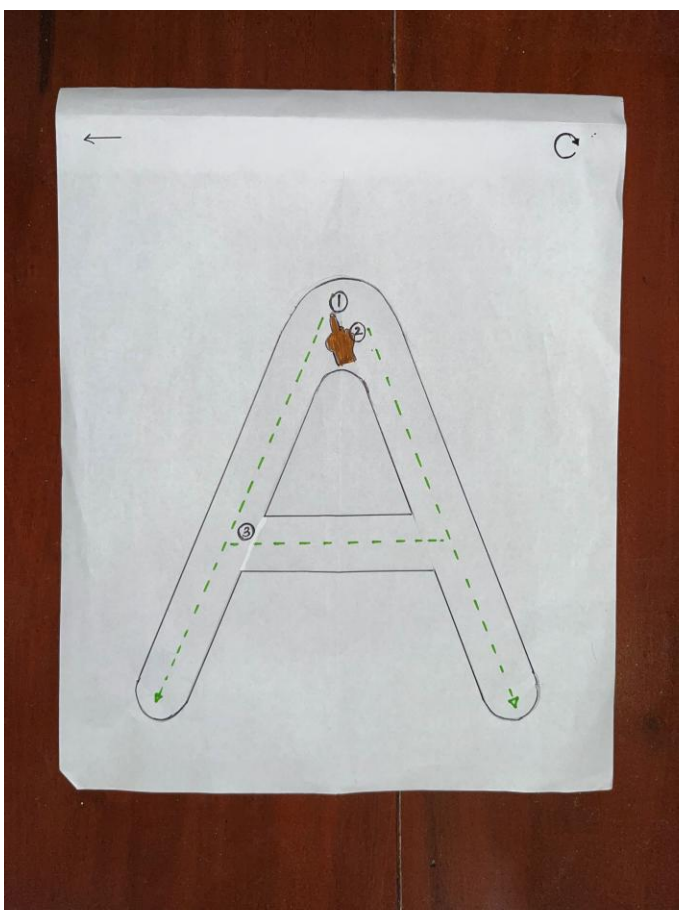

Figure 3: The Mobile Tracing Screen

6.2 Conceptual Models

- Game architecture diagram showing the distributed PC-Firebase-Mobile communication pipeline, Steam session management layer, and event logging module.
- Interaction design model mapping in-game trigger events (chest discovery) to corresponding mobile task activations and completion callbacks, illustrating the synchronized dual-device workflow.
- User flow diagrams tracing a complete play session from device pairing through chest discovery, mobile tracing task, word formation, and collective reward, illustrating the moment-by-moment handoff between PC and mobile and the points at which anticipatory feedback is triggered.

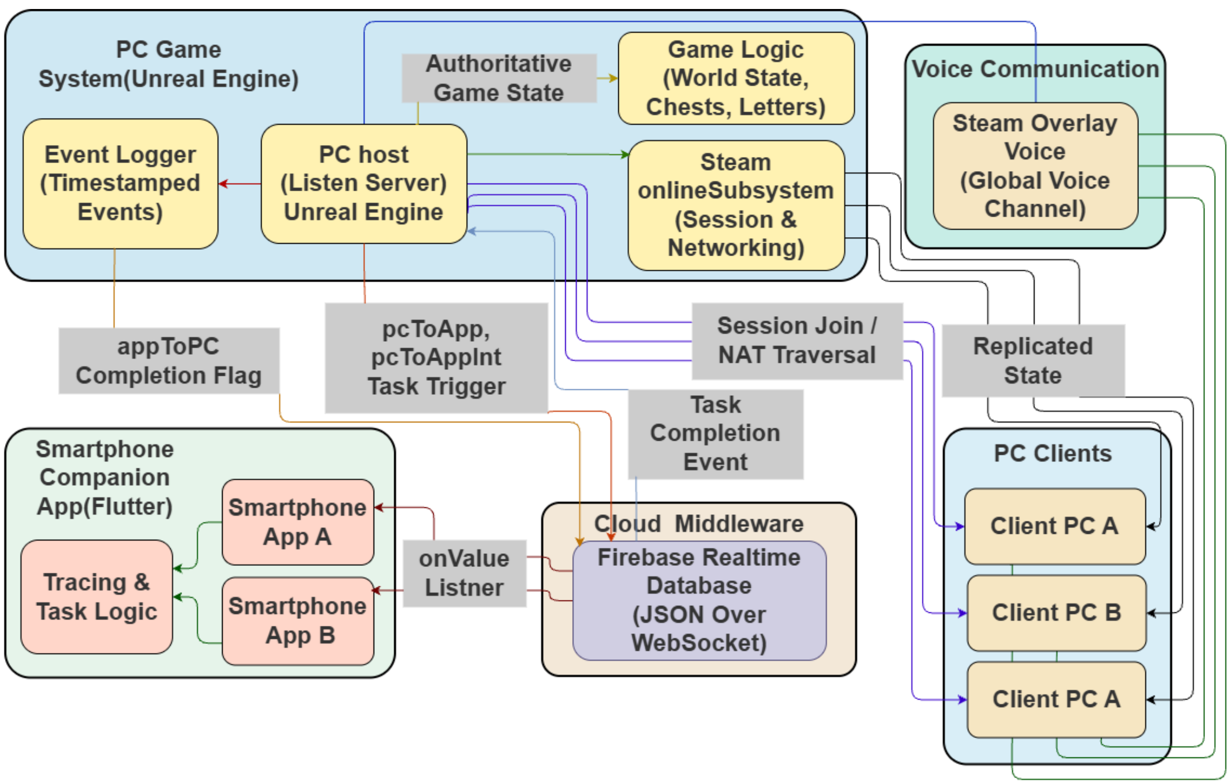

Figure 4: Game Architecture

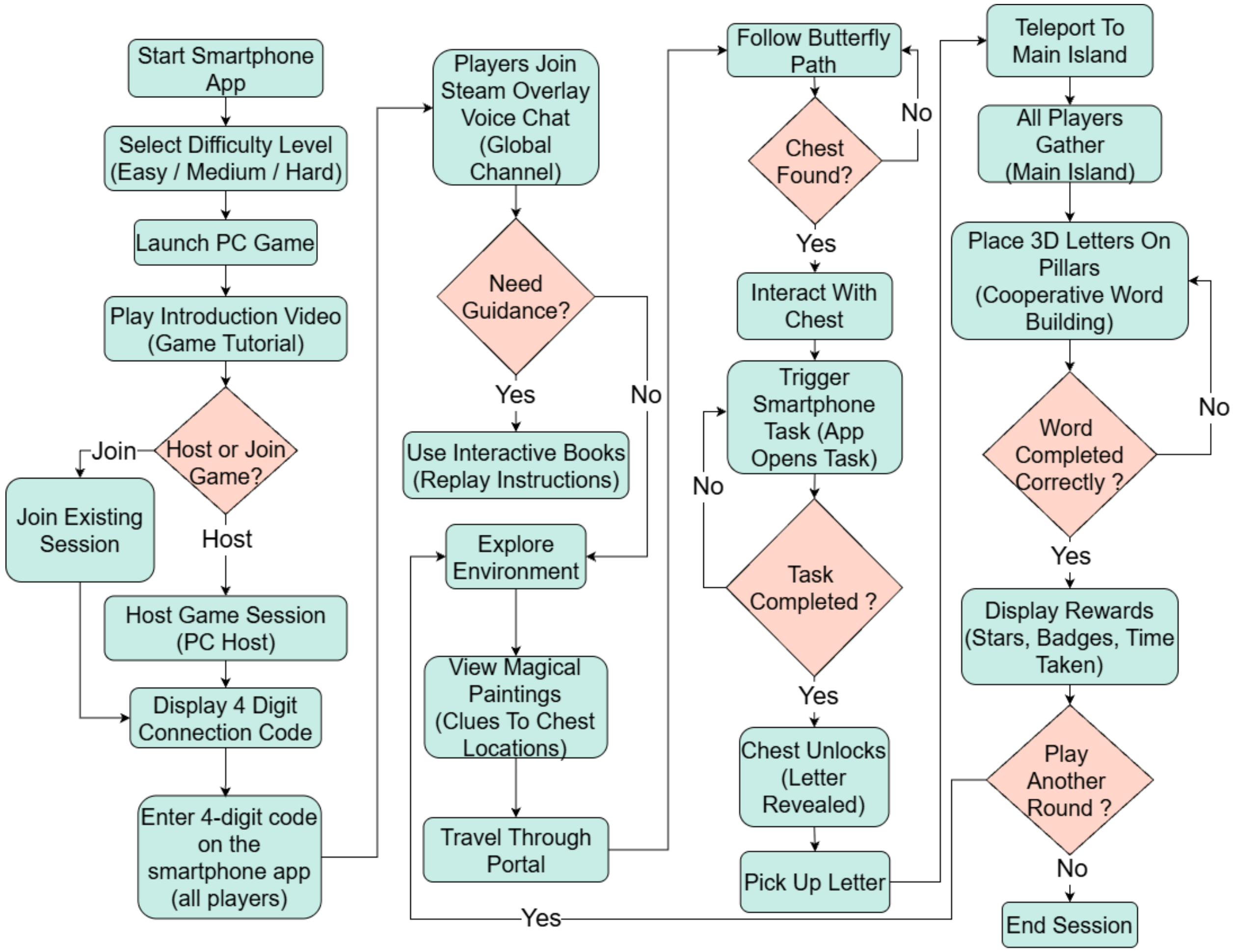

Figure 5: User Flow

6.3 Expert Evaluation of UI/UX Design

To validate the UI/UX design, mockups and a working prototype were reviewed through iterative consultation with two consultant child and adolescent psychiatrists affiliated with a pediatric hospital, selected for their specialized expertise in developmental disorders, learning disabilities, and evidence-based educational interventions. Multiple rounds of structured feedback sessions were conducted. Experts were first given comprehensive gameplay demonstrations and detailed system walkthroughs, after which their feedback was incorporated into design refinements and subsequent sessions reviewed the modifications.

Structured feedback covered emotional safety mechanisms, developmental appropriateness for children aged 8-12 with dyslexia, suitability of cooperative mechanics, alignment with evidence-based literacy intervention principles, and feasibility for supervised educational contexts. As shown in Figure 2 below, all six evaluated design dimensions met or exceeded the 4.0 threshold on a 5-point scale. Multisensory Integration and Cooperative Mechanics received perfect scores of 5.0. Emotional Safety Design, Visual Design, and Dual-Device Architecture each scored 4.5. Implementation Feasibility scored 4.0, confirming the system is technically realistic. Key qualitative findings are summarized in the table below.

Dimension	Expert Feedback
Emotional Safety	Low-visibility interaction and non-punitive feedback reduce performance anxiety
Age Suitability	Appropriate for ages 8-12 with facilitator support
Visual Design	Dyslexia-friendly colors, fonts, and 3D letters viewed favorably
Cooperative Mechanics	Shared goals without competition align with therapeutic principles
Dual-Device Architecture	Balances collaborative engagement with private practice
Multisensory Integration	Visual, auditory, and tactile modalities align with evidence-based approaches
Implementation	Requires facilitator training and session duration limits

Table 1: Expert Feedback

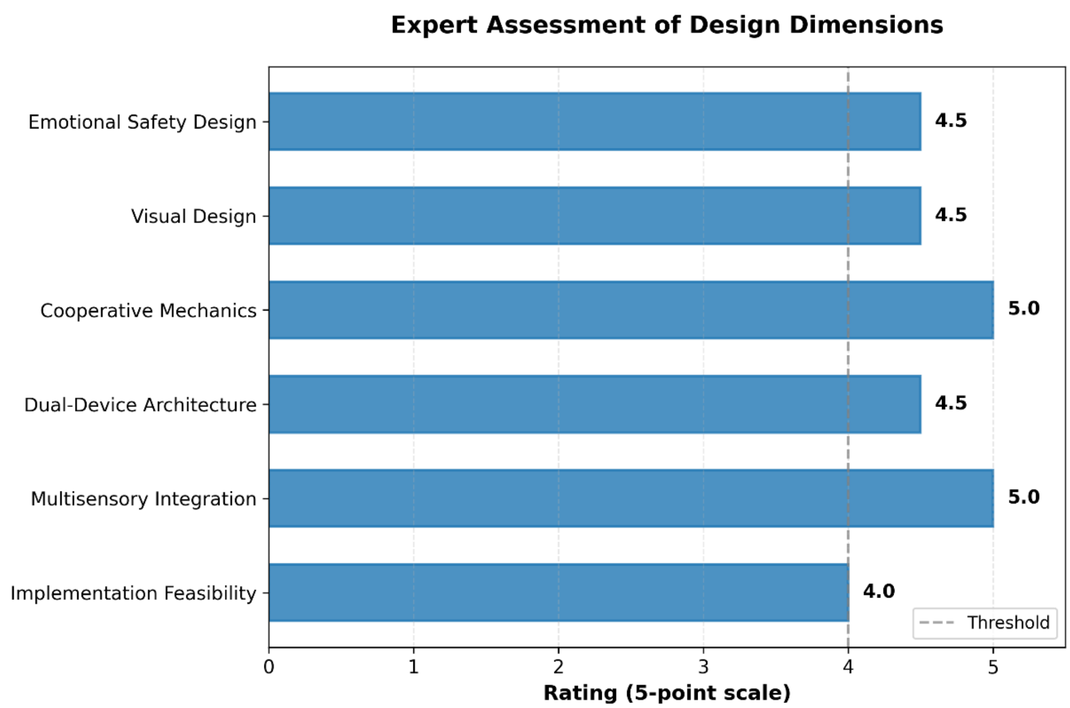

Figure 6: Expert Ratings

6.4 Initial Changes

Following iterative expert consultation, several design changes were implemented before finalising the prototype. All text-based instructions in the mobile application were replaced with icon-based guidance to minimize reading demands for dyslexic users. Tracing guide paths were widened and color-contrast was increased in line with dyslexia accessibility standards. In-game reward animations were redesigned to reflect collective team progress rather than individual performance, removing any competitive framing. The voice-communication toggle was repositioned as a persistent on-screen button rather than a keyboard shortcut, so children can activate it without interrupting gameplay flow. The session-code pairing flow was simplified to a single four-digit numeric entry to reduce onboarding friction. Additionally, experts emphasized the need for facilitator training requirements and session duration limits to manage screen time, and these were incorporated into the implementation guidelines for supervised deployment in schools and clinics.

6.5 Finalized UI Mockups

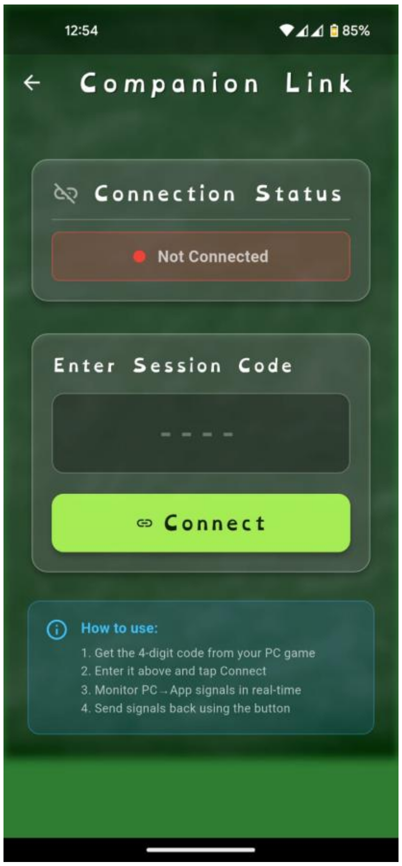

Figure 7: Session Pair screen Task

Figure 8: Customizable Settings

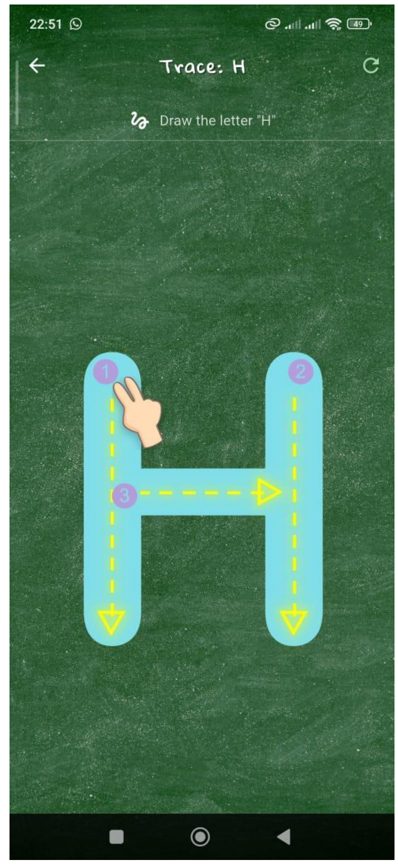

Figure 9: Letter Tracing

Figure 10: Main Game World

Figure 11: Chest Hunting

Figure 12: Word Formation Panel

Figure 13: Reward System

6.6 User Feedback / Heuristic Analysis

Heuristic evaluation was conducted by two consultant child and adolescent psychiatrists with specialized expertise in developmental disorders and learning disabilities. Both the PC game environment and the Flutter mobile application were assessed across established usability and accessibility criteria including visibility of system status, error prevention, minimal cognitive load, and consistency of visual language.

Evaluators confirmed that the low-visibility interaction design, where players are not visually exposed to one another during literacy tasks, directly reduces performance anxiety for the target age group. The non-punitive feedback approach, treating incorrect attempts as exploratory iterations rather than failures, was identified as appropriate and therapeutically aligned. The dyslexia-friendly visual choices including high-contrast colors, sans-serif typography, and 3D letter representations were viewed favorably. Cooperative mechanics built around shared goals without competitive ranking were confirmed to align with therapeutic principles for this population.

Constructive feedback highlighted two usability considerations for implementation: facilitator training is required before supervised sessions, and session duration should be actively managed to address screen time concerns for children aged 8-12. Both recommendations have been incorporated into the deployment guidelines for school and clinic settings. No critical usability failures were identified across either platform, and the system was confirmed as developmentally appropriate for the target age group prior to child user trials.

6.7 Demonstration of the App

A demonstration scenario will show two participants seated at separate PCs, each paired with an Android device via session code. Players enter the 3D game world, navigate to a letter chest, and trigger the corresponding tracing task on their mobile companion apps simultaneously. On successful completion, the chest opens and the collected letter appears in the shared word-formation panel. Players use voice chat to coordinate letter arrangement, form a target word collaboratively, and receive the collective reward animation. The demonstration highlights all four game components working in unison: multimodal gameplay, social engagement via voice, anticipatory feedback during tracing, and cognitive load distribution across devices.

Technical evaluation over three weeks with 2-5 concurrent players confirmed system feasibility. The dual-device architecture achieved over 98% task delivery accuracy between PC and mobile, with PC-to-smartphone event latency of 160-190 ms, well within acceptable range for non-real-time literacy activities. Network latency scaled predictably from 60 ms with 2 players to 150 ms with 5 players. Zero crashes were recorded across 48 hours of continuous testing, confirming architectural stability under realistic session conditions.

6.8 Future development

- Expansion of the word bank and difficulty progression to cover full primary-school literacy curricula, supporting use across multiple academic years.

- Integration of an AI-driven adaptive engine that personalises task sequencing and feedback intensity based on individual performance data collected across sessions.

- Extension to iOS and tablet platforms to broaden device accessibility, particularly for schools with iPad-based learning environments.
- Adaptation of the framework for other neurodevelopmental conditions (ADHD, dyscalculia) and for different languages, extending impact beyond English-medium dyslexia support.
- A teacher/therapist dashboard for monitoring session data, tracking individual and group progress, and adjusting difficulty settings remotely without interrupting gameplay.
- Scaling pathways under active consideration include: a formal research partnership with Lady Ridgeway Hospital and Sirimavo Bandaranaike Specialized Children's Hospital for a longitudinal clinical trial post-graduation; grant applications to the National Science Foundation of Sri Lanka and international EdTech funding bodies; and an open-source release of the design guidelines and interaction framework to enable adoption and further research by the global assistive education technology community.

7. Impact

This game creates measurable impact across three dimensions. For individual learners, it provides a safe, motivating space where dyslexic children can practice literacy skills alongside peers without fear of public failure, addressing both the cognitive and the emotional barriers that traditional tools ignore. The anticipatory feedback and cooperative task structure are designed to build genuine confidence and persistence, not merely task completion rates.

For the research community, the project generates the first controlled empirical evidence on synchronous multiplayer, cross-device, voice-enabled literacy gameplay for dyslexic children, filling a critical gap identified across multiple systematic reviews. The design guidelines and validated prototype produced will inform the next generation of assistive educational technologies globally.

For society, the project advances inclusive education in Sri Lanka by creating an accessible, low-cost digital tool deployable in schools and specialist clinics. Through partnerships with Lady Ridgeway Hospital and Sirimavo Bandaranaike Specialized Children's Hospital, the game has a realistic pathway to reach children who currently have no access to specialist digital dyslexia support, reducing educational inequity and improving long-term outcomes for a population that is frequently underserved.

Projected impact metrics: at least 60 dyslexic children reached directly in the initial clinical evaluation at both partner hospitals; validated design guidelines disseminated to an estimated 200+ researchers and practitioners globally via open-access publication; potential to scale to 500+ children in Sri Lankan schools within two years of clinical validation if institutional licensing is adopted by five partner schools; and estimated reduction in per-child specialist intervention cost of up to 40% compared to one-to-one therapist-led sessions, by enabling semi-supervised group use in school settings.

8. Call to Action

We invite educators, clinicians, parents, and policy-makers to support and engage with this research. Schools and therapy centers interested in participating in the evaluation phase should contact the research team through SLIIT's Department of Information Technology. Feedback from practitioners and families will directly shape the final game and the design guidelines it produces.

For the research and technology community: we welcome collaboration on extending the game to additional languages, conditions, and platforms. The modular architecture is deliberately open to expansion. Developers, researchers, and organizations working in assistive education technology are encouraged to reach out to explore joint research and commercialization opportunities.

Every child with dyslexia deserves more than isolation and repeated failure. This project is a step toward evidence-based, joyful, and socially connected literacy learning, and we need the broader community's involvement to make it a reality.

Our specific asks: (1) Funding or sponsorship to support clinical evaluation costs, participant recruitment, and post-graduation scaling. (2) Mentorship from EdTech investors or commercialization specialists with experience in assistive education markets. (3) Partnership expressions of interest from schools, therapy centers, or NGOs wishing to participate in the evaluation or pilot deployment phase. To get involved, contact the project team through the SLIIT, Faculty of Computing, New Kandy Road, Malabe, Sri Lanka, or reach the supervisory team directly via Mr. Didula Thanaweera Arachchi (didula.c@sliit.lk) or Mr. Nushkan Nisme (nushkan.n@sliit.lk) at SLIIT.

9. Video Submission Link

https://drive.google.com/drive/folders/18UNdghRFQAEG9DoFzem0Bj0Bx53uMPkx?usp=sharing

References

1. *Polychroni et al. (2024) - Study on reading self-concept and trait anxiety in dyslexic learners compared to typical peers*
2. *Tlili et al. (2022) - Systematic review on game-based learning benefits for learners with disabilities, covering engagement and social participation*
3. *Vonthron et al. (2024) - Study on usability and literacy gains using a rhythm-training serious game for dyslexic children*
4. *Vasalou et al. (2017) - Study on group game-play for dyslexic children showing spontaneous peer scaffolding and increased social engagement*

5. *Lee et al. (2018) - Study on second-screen companion applications and their effect on attention allocation under medium cognitive load*